

Week 7 — Six-Axis Benchmark

Same Week 6 train/test split (`random_state=42`), stratified on (`esi`) reused throughout, so every row below is comparing the same 11,025 held-out patients.

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	Recall ESI 1	Train time	Infer time / prediction	Explainability
Baseline (LogReg)	0.667	0.582	0.463	0.492	0.250	2.8 s	0.003 ms	High
Random Forest	0.639	0.468	0.368	0.389	0.000	25.8 s	0.093 ms	Medium
Gradient Boosting	0.550	0.410	0.547	0.416	0.313	10.6 s	0.009 ms	Low
Small MLP	0.641	0.536	0.456	0.485	0.250	86.1 s	0.011 ms	Low

Supporting runs:

- Hyperparameter-tuned Random Forest (`RandomizedSearchCV`, `f1_macro` scoring): best params (`n_estimators=100`, `min_samples_leaf=2`, `max_features="log2"`, `max_depth=None`), macro F1 **0.470** — better than the untuned forest (0.389), still below the baseline (0.492).
- Random Forest + one-hot demographics (`ethnicity`, `race`, `age`, `gender`): macro F1 **0.387** vs. **0.389** without them — a difference smaller than run-to-run noise. Adding fairness-sensitive demographic columns bought nothing here.

What this table actually says

No model beats the baseline on the primary aggregate metric (macro F1). The plain logistic regression baseline from Week 6 has the best macro F1 of the four (0.492), and the best explainability score. Every fancier model traded that away for something else — and in three of four cases, didn't even get a clear win in return.

The one real gain is Gradient Boosting's recall. It catches **31.3%** of ESI 1 patients (5 of 16 in the test set) versus the baseline's 25% (4 of 16) — a genuine, if small, improvement on the exact metric we chose as clinically primary in Week 6. Its macro-recall (0.547) is also the best of the four, meaning it's the model most willing to flag urgency across the board. That

comes at a real cost: the lowest accuracy (0.550) and precision (0.410) of any model, meaning more false alarms overall, and it is the least explainable of the four without SHAP.

Random Forest is a clear loser here. Untuned, it has the worst macro F1 *and* zero ESI 1 recall — worse than the baseline on every quality axis, for 9× the training time. Tuning recovers some of that (0.470 macro F1) but still doesn't clear the baseline.

A methodology note on the numbers above: Random Forest and the tuned search used a reduced tree count and search budget (150 trees, 4-iteration search with 2-fold CV) to fit within this environment's single-core compute — the assignment's own suggested settings are 300 trees and an 8-iteration, 3-fold search, which is what the submitted notebook actually runs. Re-running locally or on Colab with the full settings may shift these figures slightly, though the overall pattern — complexity not paying for itself except via Gradient Boosting's recall gain — is unlikely to reverse given how wide the gaps are.